SYBCA-AI (Sem-III)
303 Machine Learning
Practical Journal Submission (Mini Project)

Instructions:

- Students may work individually or in a group of two students.
- Select three project titles in order of preference. You can select the titles from given list or you can give me your own title which is suitable for your current semester's syllabus of machine learning.
- You need to select the dataset which is suitable for your project title. For that, dataset sources are listed below. You can find dataset from mentioned locations.
- Workflow for your project and file submission content is given below. You have to work accordingly.

Project Approval

Before starting:

- Topic must be approved by the faculty.
- Dataset must be approved by the faculty.
- Students cannot change the topic without prior permission.

Project titles:

A. Education

1. Student Academic Performance Prediction
2. Student Attendance Analysis and Performance Prediction
3. Student Placement Eligibility Prediction
4. Online Learning Behavior Analysis
5. College Admission Chance Prediction
6. Student Dropout Risk Prediction

B. Healthcare

7. Diabetes Prediction System
8. Heart Disease Prediction
9. Breast Cancer Detection using Machine Learning
10. Medical Insurance Cost Prediction
11. Obesity Level Prediction
12. Patient Hospital Stay Duration Prediction

C. Business & Sales

13. Advertising Sales Prediction
14. Retail Sales Forecasting
15. Customer Purchase Prediction
16. Product Demand Forecasting
17. Customer Churn Prediction
18. Bank Loan Approval Prediction

D. Finance

19. Credit Card Default Prediction
20. Personal Loan Eligibility Prediction
21. House Price Prediction
22. Car Price Prediction

E. Agriculture & Environment

23. Crop Yield Prediction
24. Rainfall Prediction
25. Air Quality Index Prediction
26. Water Quality Classification

F. Transportation & Travel

27. Used Vehicle Price Prediction
28. Taxi Fare Prediction
29. Flight Ticket Price Prediction
30. Hotel Booking Cancellation Prediction

Dataset Sources:

Students can obtain datasets from:

- Kaggle
- UCI Machine Learning Repository
- Data.gov portals
- GitHub public datasets
- OpenML

Minimum requirement of your dataset:

- Dataset should contain minimum 300 records.
- Dataset should contain 5 or more attributes/features.
- Dataset must be a real-world dataset.

Python Libraries

Students may use the following libraries:

- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- ydata-profiling (Pandas Profiling)
- Sweetviz
- AutoViz
- D-Tale

Workflow for your project:

No	Practical	Submission
1	**Project Proposal** Students submit • Project Title • Problem Statement • Dataset Source • Objective • Expected Output Example: "Student Academic Performance Prediction using Student Performance Dataset."	1 page proposal

2	**Dataset Collection** Students • Download dataset • Study attributes • Identify Target Variable • Identify Feature Variables	Dataset description
3	**Understanding Data (Unit-1)** Students identify • Numerical columns • Categorical columns • Missing values • Duplicate values	Dataset summary
4	**Exploratory Data Analysis (Unit-1 & 2)** Students perform • Histogram • Box Plot • Pie Chart • Bar Chart • Correlation Matrix	EDA Report
5	**Handle (Unit-2)** • Missing values • Outliers	Before vs After comparison
6	**Automated EDA (Unit-2)** Students use • Pandas Profiling • Sweetviz • AutoViz • D-Tale	Generated HTML report (Any one)
7	**Feature Selection** Students identify • Dependent Variable • Independent Variables • Train-Test Split	Feature selection report
8	**Build ML Model (Unit-4)** Depending on dataset • Regression Dataset → Linear Regression • Classification Dataset → Logistic Regression	
9	**Model Evaluation (Unit-3 & 4)** Regression • R^2 • MSE • RMSE Classification • Accuracy	Performance report

	• Precision • Recall • F1 Score	
10	**Improve Model (Unit-4)** Students apply • Cross Validation • Regularization • Compare results	
11	**Python Code**	Program code
12	**Results**	Results
13	**Conclusion & Future Scope** Students discuss • Findings • Future scope	Conclusion & Future Scope
14	**Softcopy Submission** • Python code • Dataset	Softcopy

Softcopy submission Folder Structure:

Project_Name_rollno_rollno (Folder)

a. Dataset (Sub-Folder)
    i. dataset.csv
b. Python_Code (Sub-Folder)
    i. project.py
c. Reports (Sub-Folder)
    i. EDA_Report.html
    ii. Project_Report.pdf

Report Format

Following are the topics, you have to follow for preparing report

1. Cover Page
2. Certificate
3. Problem Statement
4. Objectives
5. Dataset Description
6. Data Preprocessing
7. Exploratory Data Analysis
8. Model Building (Code)
9. Model Evaluation
10. Results
11. Conclusion
12. Future Scope

Use of AI Tools

- Students may use AI tools (ChatGPT, Gemini, Copilot, etc.) for guidance, understanding concepts, debugging and improving documentation.

- Copying complete projects without understanding is not acceptable. Because you will not learn anything from this mini project.

Important Note:

- This mini project is intended to help you understand the complete Machine Learning workflow.
- Students should be able to explain every step of their project, including dataset selection, data preprocessing, EDA, model building, and model evaluation.
- Faculty may ask questions during practical sessions based on your submitted work.